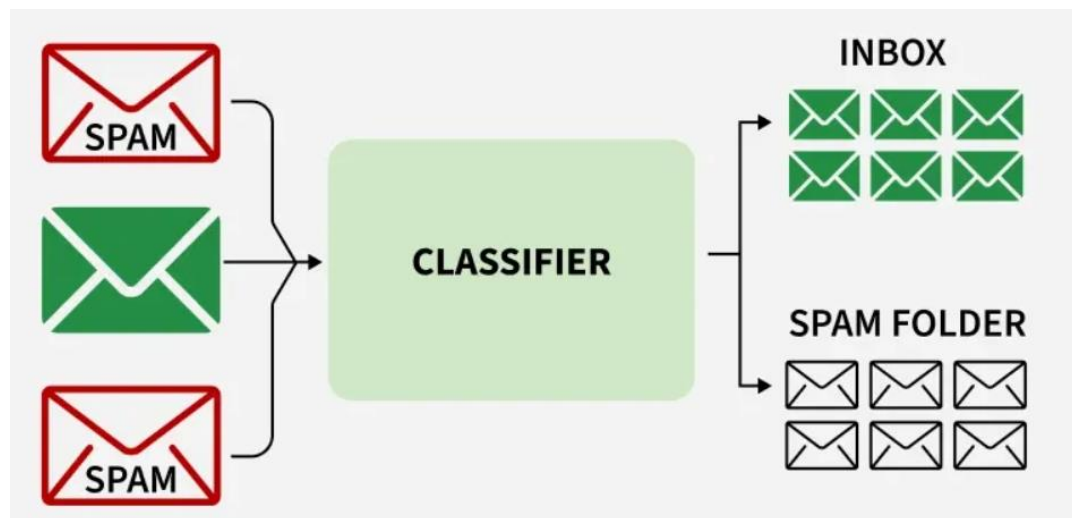


What is Text Classification?

Text Classification is the process of assigning predefined categories or labels to text data. It is a core task in Natural Language Processing (NLP) used in applications like spam detection, sentiment analysis, topic labeling, news categorization, intent detection and more. Text classification takes raw textual data and transforms it into structured information by predicting which class the input text belongs to. The general workflow involves Text Preprocessing, Feature Extraction / Representation, Model Building, Prediction and Evaluation.



Methods Used for Text Classification

1. Traditional Machine Learning Approaches: Text classification using traditional ML methods relies on algorithms like Naive Bayes, Support Vector Machines (SVM), Logistic Regression, Decision Trees, Random Forest and k-Nearest Neighbors (k-NN). These models require feature extraction techniques to convert

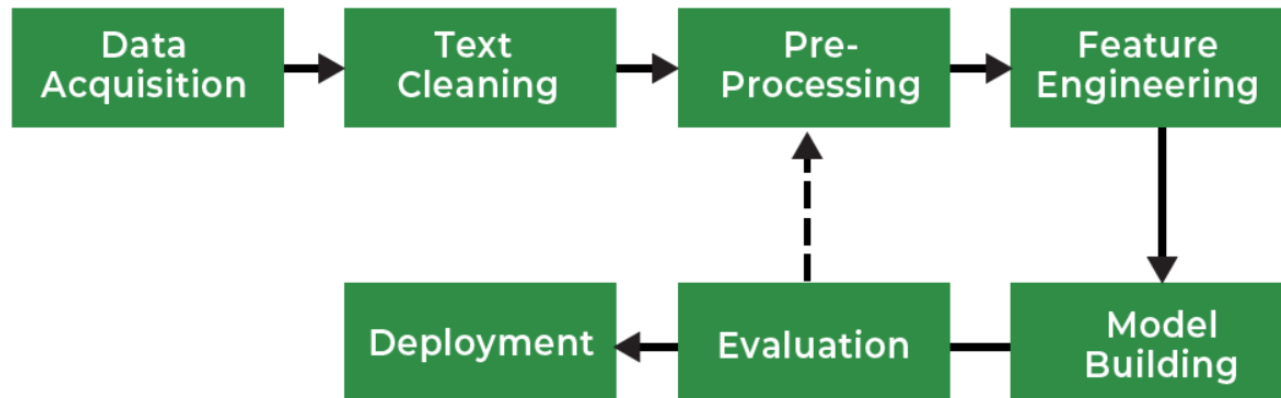
raw text into numerical form. Preprocessing steps are important. Libraries such as scikit-learn and nltk are commonly used to implement these steps.

2. Deep Learning Approaches: Deep learning methods for text classification make use of neural networks that can learn complex patterns in sequences. Models such as RNNs and CNNs are popular due to their ability to capture word dependencies and context. Libraries like TensorFlow, Keras and PyTorch provide tools to build and train these models.

3. Transformer-based Models (SOTA): Transformers represent the current state-of-the-art in text classification. Models like [BERT](#) leverage self-attention and contextual embeddings for accurate predictions. They can be fine-tuned on a specific classification task with high accuracy. Libraries such as HuggingFace's transformers provide easy access to these models. Transformer-based models are computationally expensive but significantly outperform earlier methods.

4. Zero-shot and Few-shot Classification: In zero-shot classification, models can categorize text into labels they haven't seen during training by relying on natural language inference. Few-shot methods require only a small amount of labeled data to generalize well. Models like GPT and BART enable these techniques using large-scale pretraining. Libraries like Transformers and APIs from Open AI facilitate implementation.

Illustration of Text Classification Pipeline



Basic Working of Text Classification Pipeline:

1. **Input Text:** "I love this movie!"
2. **Data Preprocessing:** Preprocessing data to gain better insights and model accuracy.
3. **Vectorization:** TF-IDF, Bag-of-Words and other techniques are incorporated for Text Representation.
4. **Model Training:** Classifier (like Naive Bayes) learns patterns.
5. **Prediction:** "Positive Sentiment" or "Negative Sentiment"
6. **Evaluation Metrics:** Accuracy, Precision, Recall, F1-Score

Applications in Text Classification

1. **Spam Detection:** Automatically classifies incoming emails or messages as spam or not spam. Widely used in emails to filter out unwanted content.
2. **Sentiment Analysis:** Identifies the emotional tone (positive, negative, neutral) of user reviews, social media posts or customer feedback. Helps businesses understand public perception of their products or services.
3. **Topic Labeling:** Assigns predefined topics or categories to news articles, blog posts or documents. Useful in content recommendation systems or content management.
4. **Language Detection:** Determines the language of a given text. Commonly used in multilingual applications and automatic translation systems.
5. **Toxic Comment Classification:** Detects abusive, offensive or toxic language in online platforms or communities. Helps in content moderation and enforcing platform policies.